

An adaptive graph neural network-based intrusion detection system for airborne network

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ABSTRACT

With the rapid development of airborne information technology, the security of airborne networks has received more attention. To provide effective protection for airborne networks, we propose an adaptive graph neural network intrusion detection system; containing a graph generation process and a graph reconstruction process. Firstly, there is inherent uncertainty in the classification boundary for imbalanced datasets in airborne networks. We propose a Weight-based k-nearest neighbor to assign differential weights, thereby constructing the adjacent weight matrix. Then, we introduce a Gaussian graph aggregation classification method to improve the accuracy of the initial graph construction. In this process, the data with connected relationships are mapped into a low-dimensional Gaussian distribution through aggregation. We sample from the Gaussian distribution for each node data to establish new connections through the latent variables for anomaly intrusion detection. Finally, the proposed method was tested on public network intrusion detection and aviation data bus intrusion detection to verify its effectiveness. The experimental results demonstrate that the proposed method improves model classification performance by using an adaptive graph neural network that repeatedly aggregates features from similar nodes, leading to more accurate detection results.

1. Introduction

With the increasing complexity of integrated avionics systems, several potential vulnerabilities have emerged, exposing aircraft to various intrusion threats (Lei et al., 2017). Therefore, ensuring the security of airborne networks has become a critical research priority. Intrusion detection systems (IDS) for airborne networks have also received increased attention in recent years.

Modern IDS use traditional machine learning and deep learning techniques to analyze the features of communication traffic, identify the patterns in collected data, and detect the malicious traffic corresponding to attacks (Mushtaq et al., 2022; Latif et al., 2024). One great challenge to face is that cybersecurity datasets are unbalanced with a few number malicious samples compared to normal class samples. Le et al. (Thi-Thu-Huong Le et al., 2024) sought to architect an innovative confluence of hybrid sampling and ensemble classification, which operated as a cohesive blend of undersampling and oversampling techniques. Li et al. (Huang et al., 2024) used the SMOTE oversampling method to generate malicious data samples so that the ratio of normal

data in the training set to malicious data in the test set is similar. However, there is a lack of research and analysis on the problem of classifying data nodes at classification boundaries, leading to inaccurate classification boundaries. On the other hand, when the intrusion data is represented as network traffic flows, these existing methods often overlook or fail to effectively exploit the inherent graph structure. To address the graph structure-based data, researchers are exploring a variety of techniques to improve the security of aircraft network systems.

Recently, graph neural network (GNN)-based methods have gained more attention in network intrusion detection due to their ability to deal with graphical structure data, such as graph convolution network (GCN) (Kipf and Wainwright, 2016) and graph attention network (GAT) (Veličković et al., 2017). GNN exploits the graph structures by using messages that pass between nodes and/or edges, allowing the neural network to effectively learn from graph-based data and generate low-dimensional vector representations (Peng et al., 2023; Cai et al., 2017; Tristan et al., 2023). Hamilton et al. proposed an inductive method called GraphSAGE, which aims to leverage node feature information to efficiently generate node embeddings for unseen data (Will et al., 2017).

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Mahsa et al. extended the GraphSAGE to introduce the E-GraphSAGE model, the difference being that this model takes into account both the characteristics of the source node and the edge (Mirlashari and Rizvi, 2024). To pay more attention to the edge information, Evan et al. presented the Anomal-E approach, which leverages edge features and a graph topological structure in a self-supervised manner (Caville et al., 2022).

In practical applications, GNN models often face the challenge of obtaining accurate graph topologies for network intrusion. An imprecise network topology can reduce the ability to accurately express node features and consequently affect the performance of GNN models. Another is that some popular learning algorithms will create k nearest neighbors (KNN) based on feature similarity among nodes (Zhang et al., 2023; Tenenbaum et al., 2000; Roweis and Saul, 2000). Each node is connected only to its top k similar neighbors. In addition, relying only on the KNN method to construct an association graph for neighbor nodes will have certain limitations. The similarity-based composition method also faces the problem of classification decisions at boundary nodes, because the data samples at the boundary nodes may have similarities with the data on both sides of the boundary. Thus framing the graph in this way still aggregates two classes of data during the learning process of the graph neural network, leading to classification difficulties.

To address the aforementioned challenges of GNN-based intrusion detection, we propose an adaptive GNN-based IDS for airborne networks. This model combines Weight-based K-nearest neighbor (W-KNN) and Gaussian graph aggregation classification, named WKG. It contains a graph generation process and a graph reconstruction process to ensure accurate intrusion detection. The main contributions of this work can be summarized as follows:

- (1) In constructing graph structures from network intrusion data, the W-KNN method is proposed to assign connection weights between nodes, with higher weights assigned to nodes that exhibit greater similarity. This weight assignment strategy can improve the classification performance at the decision boundary nodes during the subsequent graph aggregation process.
- (2) The Gaussian graph aggregation classification model is applied to improve the accuracy of the initial network construction. In this process, the node features are projected into a low-dimensional Gaussian distribution, from which latent variables are sampled. An attention mechanism is then employed to classify each node's latent variables.
- (3) We conducted experiments on two datasets, considering both the traffic data of the MIL-STD-1553 data bus and the public network UNSW-NB15-v2 in airborne networks. Our method was compared with baseline methods, and the experimental results demonstrate that the proposed method achieves higher intrusion detection accuracy.

The remainder of this paper is organized as follows. Section 2 introduces related works. Section 3 presents the proposed WKG method in detail. Section 4 describes the experiment. Sections 5 and 6 report the experimental results and discussion. Finally, Section 7 concludes closes the paper with conclusions and future work.

2. Related works

2.1. IDS for airborne network

The airborne network is a critical communication infrastructure within an aircraft, connecting numerous avionics devices and systems on the aircraft to ensure the exchange of data transfer and control signals. It mainly consists of the avionics data bus network and the in-flight Entertainment network. The MIL-STD-1553 protocol (De Santo et al., 2021; Maximilian et al., 2022), a widely used standard in avionics, uses a multipoint topology of terminals connected via a dual redundancy data

bus. Marwa et al. analyzed the data bus traffic and proposed an adaptive IDS by stacking ensemble techniques and LSTM (Elsayed et al., 2022). Inspired by the histogram comparison approach, Sebastien et al. used histograms to model the timing characteristics of the MIL-STD-1553 protocol and improved the temporal resolution of threat detection (Généreux et al., 2020). Modern IFE systems, designed to provide passengers with a range of entertainment and information services during a flight, are increasingly adopting standard Ethernet technology for its versatility and cost-effectiveness. Huang et al. proposed a hierarchical differential evolution based on IDS to identify unknown cyberattacks (Huang et al., 2023). Zoghi et al. conducted a deep insight into the intricacies of the UNSW-NB15 dataset, and the data overlap problem can be detected by the visualization method (Zoghi and Serpen, 2023). Cui et al. (2023) proposed a groundbreaking multi-module integrated intrusion detection system named GMM-WGAN-IDS, which is designed to obtain deep feature representations from high-dimensional data. Abbas et al. (2023) have put forward a novel hybrid ensemble model that employs a random forest-recursive feature elimination method, which aims to enhance the predictive performance by intelligently selecting the most relevant features.

These abovementioned methods are often hard to deal with the inherent graph structure of network traffic flows and identify anomalous nodes and behaviors. Traditional deep learning approaches tend to treat data as Euclidean constructs, thereby overlooking the non-Euclidean. In contrast, GNNs can process graph-structured data directly, preserving the topological nuances of network traffic.

2.2. Graph neural network-based method for network intrusion detection

Nowadays, GNNs have garnered intensive attention for network intrusion detection due to their suitability for representing network traffic flows (Kiran Boppana and Bagade, 2023; Wang et al., 2023). Extracting features from the raw data of network flow samples is vital in the learning process. Aiming to improve detection performance by capturing more features from data flows, Deng et al. (Deng and Huang, 2025) proposed an edge-featured multi-hop attention graph neural network for IDS. This method enhances computational efficiency through attention propagation and integrates node and edge features, fully leveraging data characteristics. To tackle the difficulty that hand-crafted features cannot be manipulated for more fine-grained operations, Lin et al. (2023) proposed a multi-modal deep learning framework that combines the self-attention mechanism, to learn the deep information between packets in the network flow.

Network flows need to be manually annotated as supervisory labels, a process that is time-consuming or even impossible. Xu et al. (2024) proposed a self-supervised graph representation learning method for identifying malicious attacks, without manual labeling or any prior knowledge. This reduces the labor and time costs while expanding the scope of intrusion detection in networks. To address privacy issues in the application of graph representation learning, Pei et al. (2024) proposed a privacy preserving graph neural network for IDS. This network constructed a private graph convolution layer to prevent adversaries from inferring sensitive information. Zhu et al. (2024) proposed a Darknet GNN for Darknet traffic classification, combining Darknet traffic graph and attention mechanism. The graph structure captures the interactions between local clients and remote servers in darknet traffic, which can effectively curb malicious darknet activities.

3. Proposed method

In airborne network traffic flows, the structured graph is a collection of vertices and edges. Let $G(V, E, W)$ denote an undirected graph, with node V , edge E , and edge weight W . For node $v_i \in V$, the elements $e_{ij} = (v_i, v_j) \in E$ (Mei et al., 2023; Han et al., 2022), and $w_{ij} \in W$ denote the edges and weights of the edges of two nodes v_i and v_j , respectively.

Usually, an association graph with N nodes can be represented by a weight adjacency matrix $A \in R^{n \times n}$, indicating the topological structure of the graph, when nodes i and j are linked, $A_{ij} = w_{ij}$, otherwise $A_{ij} = 0$. Then the Laplacian graph L can be obtained by $L = D - A$, where D is the degree matrix of A .

3.1. The proposed WKG model

This section introduces the proposed WKG intrusion detection model, as shown in Fig. 1. The model mainly consists of two stages: Firstly, for network intrusion data, construct initial graph structures based on feature similarity. Then, to improve intrusion detection performance, a Gaussian graph aggregation classification is applied to perform feature aggregation on the constructed initial graph structure.

The first stage is to construct the initial graph structure. Firstly, we traverse all the data in the network dataset using the W-KNN method to find the k nearest neighbors for each data node. Then, the exponential decay method is applied to assign weights to the k data nodes based on their similarity, with the more similar data node receiving the higher weight. The purpose of this is to assign larger aggregation weights to the data nodes with greater similarity during the graph aggregation process. Finally, the adjacency weight matrix A is constructed for the dataset based on the W-KNN lookup results and the assigned weights.

The second stage is the Gaussian graph aggregation classification. Firstly, the constructed graph structure data can be put into the GCN to learn the graph structure features. In this process, each data node aggregates the features of its k most similar nodes based on the connection weights. Then, we use the KL divergence as a constraint to map the aggregated features to a Gaussian distribution. The latent variable Z , obtained by sampling from the Gaussian distribution, can be used to reconstruct the graph structure. Therefore, we use the reconstruction error as a constraint for graph reconstruction learning. Finally, we perform aggregation learning on the reconstructed graph structure data to classify each node. It is worth noting that the data used in the secondary aggregation learning is the latent variable Z .

3.2. Graph generator W-KNN

Traditional similarity-based methods have challenges at classification boundaries because nodes near the boundary may have similarities with both classes, leading to inaccurate classification of boundary nodes. The W-KNN method is proposed to improve the accuracy of graph construction, which adds weight constraints to the connections between data nodes during graph structure generation. As shown in Fig. 2, blue nodes represent normal data, yellow nodes represent anomaly data, and red nodes indicate decision boundary points. This method constructs graph-structured data by connecting data nodes with high similarity, which facilitates the aggregation of features from similar data nodes for classification. By assigning weights based on similarity, the model will focus more on the class with the higher weight, thereby reducing the misjudgment of boundary nodes.

Assuming the samples in the airborne network dataset are denoted as $X = \{x_1, x_2, \dots, x_n\}$, the Euclidean distance between each pair of data nodes x_i and x_j is calculated as:

$$d_{ij}(x_i, x_j) = \left(\sum_{l=1}^m |x_{i,l} - x_{j,l}|^2 \right)^{\frac{1}{2}} \quad (1)$$

where m represents the length of the sample data. In the process of constructing the association matrix, we select each node as the center, calculate the k -nearest nodes, and then assign connection weights to these k nodes based on their distances. Suppose the i th node is chosen as the center, then the weight of the j th node to the i th node is calculated as follows:

$$w_{ij}(x_i, x_j) = e^{-\alpha d_{ij}} \quad (2)$$

Where α is a learnable parameter greater than 0, d_{ij} is the Euclidean distance between two nodes.

In this paper, we use the W-KNN method to construct graphs for airborne networks. At this moment, each sample is no longer an independent instance waiting to be classified but becomes a node in the graph. The neighbor relationships determined by W-KNN will define the edges and weights between the nodes in the graph. Each edge has an

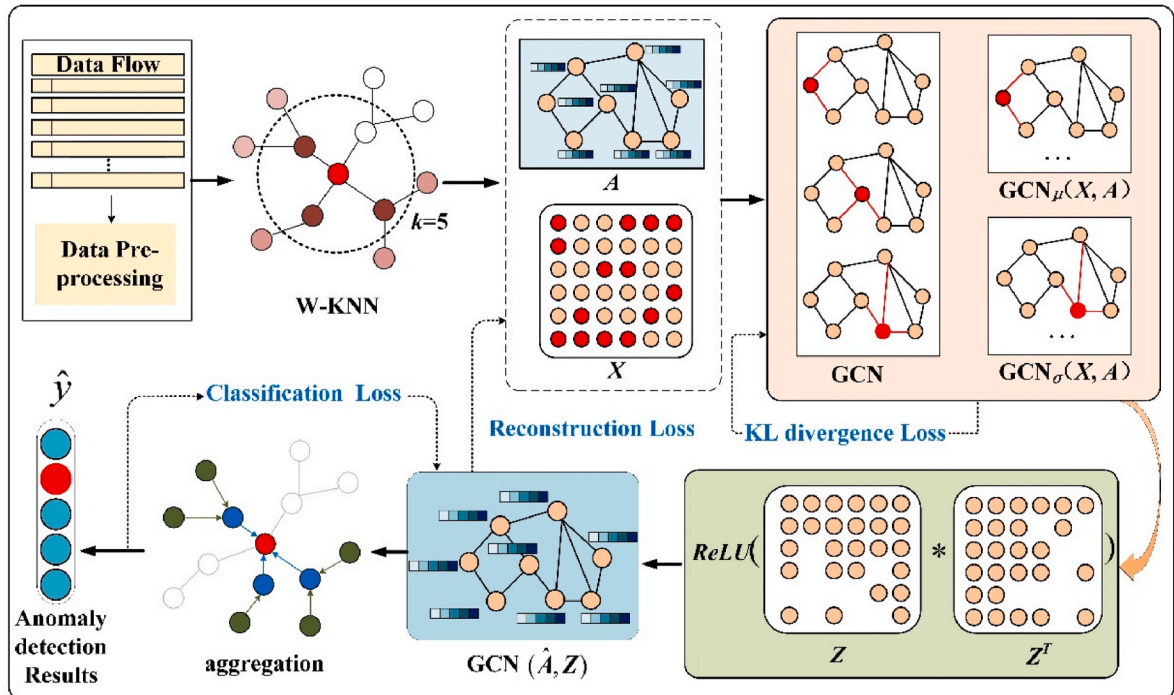


Fig. 1. The framework of WKG.

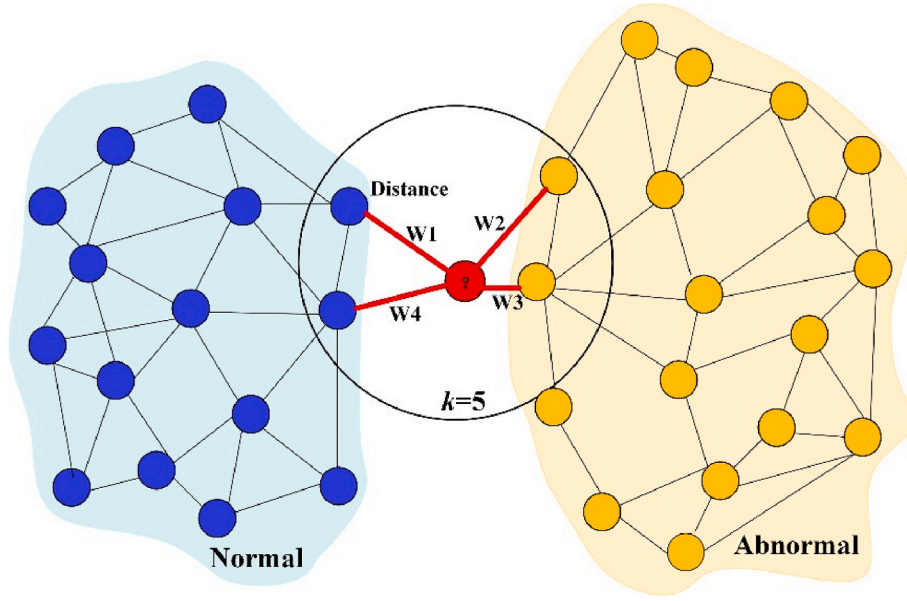


Fig. 2. Based on W-KNN graph construction.

associated weight W_{ij} , which represents the degree of similarity between the connected nodes. By combining the edges and their associated weights, a weighted adjacency matrix is constructed.

The weight range of the adjacency matrix is typically $[0,1]$, normalizing the weight function.

$$w_{ij} = \frac{e^{-ad_{ij}}}{\sum_{n=1}^k e^{-ad_{i,n}}} \quad (3)$$

The connection weights between the node i and the node j in the adjacency matrix are as follows:

$$A_{ij}(x_i, x_j) = \begin{cases} w_{ij}, & \text{if } x_j \subseteq x_i \\ 0, & \text{else} \end{cases} \quad (4)$$

Where w_{ij} is the connection weight between two nodes, and the two nodes are not connected, the weight is 0.

3.3. Gaussian graph aggregation classification

To improve the accuracy of the initial graph construction, we propose a Gaussian graph aggregation classification method to reconstruct the graph structures. The process of graph reconstruction is shown in Fig. 3. Firstly, based on the graph structure constructed in Section 3.1, the training and testing datasets are built by incorporating label information. The Gaussian graph aggregation classification is then used to reconstruct the graph structure data and perform feature classification. In this step, we use the GCN model to map the graph-structured data to a

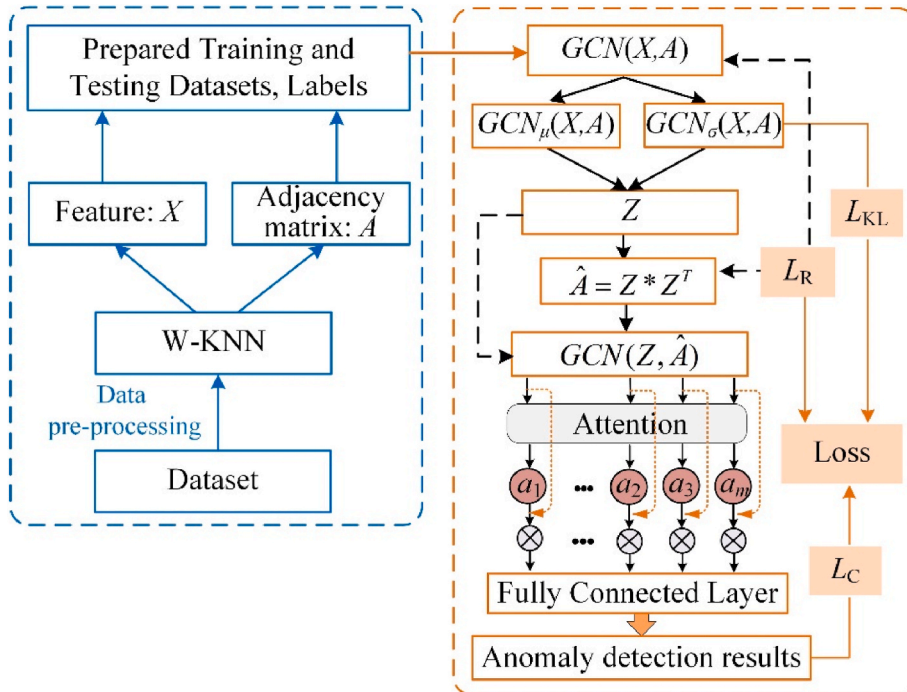


Fig. 3. Graph reconstruction process.

low-dimensional Gaussian distribution. Samples are drawn from the Gaussian distribution to obtain the data Z , which is subsequently used for graph reconstruction and feature classification. The loss function consists of L_{KL} for Gaussian distribution construction, L_R for graph reconstruction, and L_C for feature classification. It is worth noting that, during feature classification, we again apply the GCN model to perform node aggregation on the sampled data and the reconstructed adjacency weight matrix. Finally, a point attention mechanism is applied to classify each data node.

To present the Gaussian graph aggregation process more clearly, as shown in Fig. 4, we take node 5 as an example to visualize the aggregation process. Let the feature of node 1 be x_1 , and the connection weight between node 5 and node 1 be w_{51} . In the graph aggregation process, assuming that node 5 is connected to nodes 1 to 4, the graph aggregation feature of node 5 is the average of the product of the five and the weight. Note that in the graph aggregation process, each node aggregates the features of its neighboring connected nodes. After the feature aggregation is completed, a two-layer GCN model is used to compute the mean and variance for each node, which follow a normal distribution.

Firstly, the connections between nodes and the connection weights are derived from Eq. (4) in Section 3.1. During the aggregation process, node 5 is connected to nodes 1 through 4. Therefore, in the GCN (X, A) convolution process, node 5 aggregates the feature information from the other four nodes. The specific calculation process is as follows:

$$\begin{cases} GCN(X, A) = \text{ReLU}(\tilde{A}XW_0) \\ \tilde{A} = D^{-\frac{1}{2}}AD^{-\frac{1}{2}} \end{cases} \quad (5)$$

where D is the degree matrix of A , and W_0 is the weight matrix of the GCN. In GCN, each node can aggregate the features of its neighbors, thus enhancing the node information. Before this, W-KNN has assigned node connection weights $w_{i,j}$ to the adjacency matrix A .

After that, we use GCN to map the node features to a low-dimensional Gaussian distribution. This graph neural network model consists of two layers of GCN, the first layer is used to extract generic graph aggregation features as shown in Eq. (5). The second GCN layer learns the mean and variance of the feature data from the features. The mean and variance are given as follows:

$$\begin{cases} \mu = GCN_\mu(X, \tilde{A}) \\ \log \sigma = GCN_\sigma(X, \tilde{A}) \\ GCN_\mu(X, \tilde{A}) = \tilde{A} \text{ReLU}(\tilde{A}XW_0)W_\mu \\ GCN_\sigma(X, \tilde{A}) = \tilde{A} \text{ReLU}(\tilde{A}XW_0)W_\sigma \end{cases} \quad (6)$$

Where W_μ and W_σ are the trainable weight filters for GCN the layer, which share the first-layer parameters W_0 . This means that we again use GCN to get to calculate the variance and mean specific GCN model.

Then, we obtain the mean and variance of the Gaussian distribution, and sample from it to acquire the latent variables for each node. For example, the potential representation of node i can be obtained as follows:

$$q(Z_i|X, \tilde{A}) = N(Z_i|\mu_i, \text{diag}(\sigma_i^2)) \quad (7)$$

Where Z_i is the latent representation of the i th node sampled from the Gaussian distribution (μ_i, σ_i^2) . The total latent variables are calculated as follows:

$$q(Z|X, \tilde{A}) = \prod_{i=1}^N q(Z_i|X, \tilde{A}) \quad (8)$$

The graph reconstruction is given by an inner product between latent variables z_i and z_j , expressed as

$$\begin{cases} p(\hat{A}|Z) = \prod_{i=1}^N \prod_{j=1}^N p(\hat{A}_{ij}|z_i, z_j) \\ p(\hat{A}_{ij} = w_{ij}|z_i, z_j) = \sigma(z_i^T, z_j) \end{cases} \quad (9)$$

where the weights are calculated by W-KNN. If new connections are introduced during the reconstruction process, the weights are recalculated and reassigned accordingly.

The WKG model can learn a new and better graph structure $(\hat{A}, Z)$, with adjacent matrix $\hat{A}$ provided by GCN and latent representation Z provided by graph reconstruction. We use GCN again for graph aggregation and classification of potential feature Z , and the calculation formula is as follows:

$$H_{out} = GCN(Z, \hat{A}) \quad (10)$$

Where $H_{out} = \{h_1, h_2, \dots, h_n\}$ is the result of the aggregation of the features of each node.

We classify the node h_i using the point attention mechanism, calculated as follows:

$$\begin{cases} \alpha_i = \text{softmax}(h_i, q) \\ y_i = \sum_{j=1}^n \alpha_{ij}f_j \end{cases} \quad (11)$$

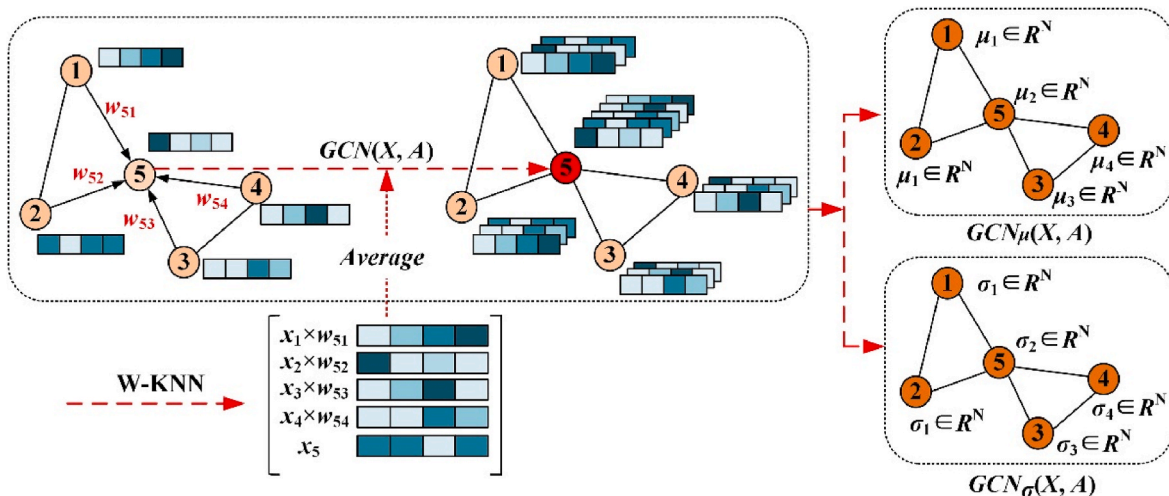


Fig. 4. The process of Gaussian graph aggregation classification.

where q is a set of learnable parameter vectors. The probability of the i th input vector is selected, and n is the number of vectors.

3.4. The WKG model training

During training, the WKG model is optimized using a combination of graph reconstruction loss, classification loss, and KL divergence loss.

Graph reconstruction is facilitated by the Gaussian graph aggregation classification to assess the discrepancy between the model's latent space distribution and the prior distribution. The Reconstruction loss is expressed as

$$L_R = E_{q(Z|X,A)}[\log p(A|Z)] \quad (12)$$

The KL divergence loss helps prevent the model from learning overly complex features, thereby enhancing its generalization capabilities through regularization, which is formulated as

$$L_{KL} = KL(Q(z|x) \| P(z)) \quad (13)$$

Where the $Q(z|x)$ is the latent space distribution, and $P(z)$ is the prior distribution.

For the classification task, a fully connected layer is deployed, and the cross-entropy loss is adopted to optimize the predictions. The Classification loss is defined as

$$L_C(\bar{y}, y) = -\bar{y}[y] + \log \left(\sum_j e^{\bar{y}[j]} \right) \quad (14)$$

Where $\bar{y}$ is the predicted label, the y represents the real label, and $\bar{y}[j]$ is the j th of $\bar{y}$.

In this research, we aim to minimize the overall network loss by concurrently optimizing for reconstruction accuracy and task-specific prediction performance. The total loss is expressed as

$$L_{Total} = L_R + L_C + L_{KL} \quad (15)$$

Algorithm 1. outlines the training process of our proposed WKG.

Algorithm 1 Pseudocode for the proposed WKG

Input: $X = \{x_1, x_2, \dots, x_n\}$ with n input data
Parameters: Batch B , epoch E
Require: Initialize $e = 0$, $b = 0$
for each $x_i \in X$ **do**
 Select k nearest neighbors $X_k = \{x_{k,1}, x_{k,2}, \dots, x_{k,k}\}$ based on Eq. (1)
 Establish the edges $V_k = \{v_{k,1}, v_{k,n}, \dots, v_{k,k}\}$ of x_i and X_k
 for $v_i \in V_k$ **do**
 Calculate the weight of edges based on Eq. (3)
 end for
 Calculate the adjacency weight matrix based on Eq. (4)
end for
Output: $G = (V, E, W)$ with n input nodes, $A \in \mathbb{R}^{n \times n}$
for $e < E$ **do**:
 for $b < B$ **do**:
 Get batch data from X and A
 Calculate the graph convolution of the first layer image using Eq. (5)
 Calculate the σ and μ using for Eq. (6)
 Sampling from σ and μ to obtain Z
 Graph reconstruction based on Z to obtain $\hat{A}$
 Graph aggregation based on $H_{out} = GCN(Z, \hat{A})$
 for $h_i \in H_{out}$ **do**
 Node classification based on point attention mechanism
 end for
 $b \leftarrow b + 1$
 end for
 $e \leftarrow e + 1$
end for

4. Experiment

4.1. Datasets and pre-processing

To verify the performance of the IDS framework based on the WKG model, this study applies it to two datasets derived from aircraft airborne networks. One is the NF-UNSW-NB15-v2 dataset representing public network traffic and the other is the MIL-STD-1553B (1553B) aviation data bus dataset. Further details on each dataset are as follows:

(1) NF-UNSW-NB15-v2

NF-UNSW-NB15-v2 NetFlow formatted version of UNSW-NB15 (Moustafa and Slay, 2015) with 43 standardized features. Out of the 2,390,275 flows, 2,295,222 (96.02 %) samples are benign flows, and the remaining 95,053 (3.96 %) samples are attack flows (Mohanad et al., 2021; Nguyen and Kashef, 2023). We randomly downsampled to 3 % due to the large size of this dataset (Xu et al., 2024). In the training phase, a substantial 80 % of the data was sampled for training, while the remaining 20 % was used for testing and performance evaluation.

(2) MIL-STD-1553B

The MIL-STD-1553B (1553B), a military integration standard, defines the communication bus used for transmitting data between the bus controller and the remote terminal. In Fig. 5, the 1553B communication traffic dataset is produced by Yahalom et al. (Ran et al., 2019), which simulated the malicious RT spoofing attacks on the 1553B system by injecting data packets. The generated RT spoofing attack data includes non-sequential datasets and sequential datasets. The 1553B dataset consists of 14,942 normal samples and 1690 abnormal samples (Onodueze and Josyula, 2020; Li et al., 2024).

4.2. Evaluation metrics and training

Performance or evaluation metrics are indicators used to evaluate the model at various stages, such as accuracy, precision, recall (also called TPR), and F1 score (Negar Mohammadreza et al., 2024; Mohanad et al., 2022). In addition, the False positive rate (FPR) and the True Positive Rate (TPR) (Li et al., 2022) are an important metric to reflect the actual performance of WKG for binary classification.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (16)$$

$$Precision = \frac{TP}{TP + FP} \quad (17)$$

$$Recall = \frac{TP}{TP + FN} \quad (18)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (19)$$

$$FPR = \frac{FP}{FP + TN} \quad (20)$$

where: True Positive (TP): the number of samples correctly identified as positives. False Positive (FP): the samples are incorrectly identified as positives when they are actually negatives. True Negative (TN): the number of samples are correctly identified as negatives. False Negative (FN): the samples are incorrectly predicted as negatives when they are actually positives. FPR: the negative samples are incorrectly classified as positive. TPR (also called recall): the actual positive samples are correctly identified.

In this paper, all the experiments are carried out on a PC with Intel Core i5-13400F CPU, 16-GB RAM, and GeForce RTX 3060 GPU. Our

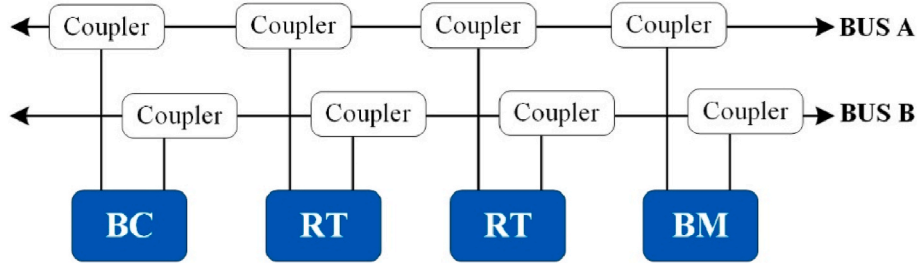


Fig. 5. The MIL-STD-1553B architecture.

model is implemented using Python and Pytorch. In addition, the detailed information on hyperparameters is provided in Table 1.

5. Anomaly detection results

5.1. The experimental results of both datasets

In Table 2, we present a comparative analysis of the performance evaluation metrics across different methods applied to the datasets of NF-UNSW-NB15-V2 and MIL-STD-1553. We evaluate the proposed method, ML methods, and GNN-based approaches, such as GCN, GAT, and GraphSAGE (GS). The performance metrics used for comparison are accuracy (A), precision (P), recall (R), and the F1 score.

In this experimental result of NF-UNSW-NB15-V2, the WKG model has a high accuracy, reaching more than 97.4 %. Especially, the GS method is better than GAT and GCN in accuracy, but it has a lower F1 than Random Forest. Compared with others, our method achieves better values in all four metrics and outperforms the NF-UNSW-NB15-V2 dataset. Overall, in the intrusion detection process of the 1553B dataset, each method can achieve outstanding results for various performance metrics. In particular, LSTM, GCN, as well as GS models have high accuracy which can successfully identify abnormal network intrusion traffic. However, it is important to note that the 1553B dataset is unbalanced with a relatively small number of samples, which impacts the performance metrics. Particularly, the SVM and GAT models do not perform well in terms of F1 score, reflecting challenges in handling unbalanced data effectively. As a result, most methods demonstrate certain limitations, struggling to perform well simultaneously on both datasets. This highlights the inherent challenges in adapting the same model across different types of data environments. Nonetheless, the results of our proposed model are significantly better than those of traditional machine learning models in most cases, indicating its robustness and effectiveness in handling diverse datasets.

To evaluate the effectiveness of our method, we performed significance analyses on both datasets using a *t*-test. When comparing our method with each traditional method, the calculated probability values (*p*) were as follows: SVM (*p* = 0.03), KNN (*p* = 0.02), LSTM (*p* = 0.01), RF (*p* = 0.04), GCN (*p* = 0.02), GAT (*p* = 0.01), and GS (*p* = 0.02). With a significance level set at 0.05, we can conclude that our model outperforms the traditional methods in terms of performance and that all these differences are statistically significant.

Table 1
Hyperparameter.

Hyperparameter	Value
<i>k</i> value	<i>k</i> = 5
Training set: Test set	8:2
Epochs	150
GCN hidden layer dimension	32
Optimizer	Adam
Learning rate	8e-4

5.2. Performance evaluation of the WKG model

The plot of accuracy and loss of the proposed model is plotted in Fig. 6 on NF-UNSW-NB15-V2. We can see in the accuracy plot that the model is well trained as the curves are now reaching a plateau in its learning process. The model has comparable performance on both train and test datasets and loss decreases when we move to a higher number of epochs.

In addition, the loss of the WKG model and the changes in the various components are shown in Fig. 7. We can see that the total loss of the WKG model decreases with epochs, indicating that the model is learning and improving its performance. The classification loss decreases with epochs, which measures the ability of the WKG model to correctly classify the data. The decreasing reconstruction loss suggests that the model is effectively learning to reconstruct graphs by simplifying their structure and reducing redundancy. The KL divergence loss decreases in the initial epochs and stabilizes at a small value, ensuring that the model ultimately converges to a good solution. In summary, the decreasing losses and their eventual stabilization indicate that the WKG model is not only becoming better at classifying and reconstructing graph data but also at learning a well-structured and regularized latent space.

As illustrated in Fig. 8, the accuracy varied with the number of iterations on the 1553B dataset, including the LSTM, GCN, GAT, GS, and the proposed method. For other methods not shown in the figure, such as the SVM, KNN, and RF, the accuracy remained unchanged as the number of iterations increased. In this figure, the accuracy of the GS, GAT, GCN, and LSTM varies with the number of iterations and can be stable after 40 epochs. The WKG model shows a different pattern where its accuracy continues to improve until it levels off after 60 epochs. Notably the WKG model outperforms the other models after reaching a plateau, highlighting the advantage of the WKG model for graph reconstruction. By benefiting from the extended training, the WKG model can deeply learn the network structure from the complexity of the dataset, effectively capturing and exploiting the intricate connections and patterns inherent in the 1553B dataset.

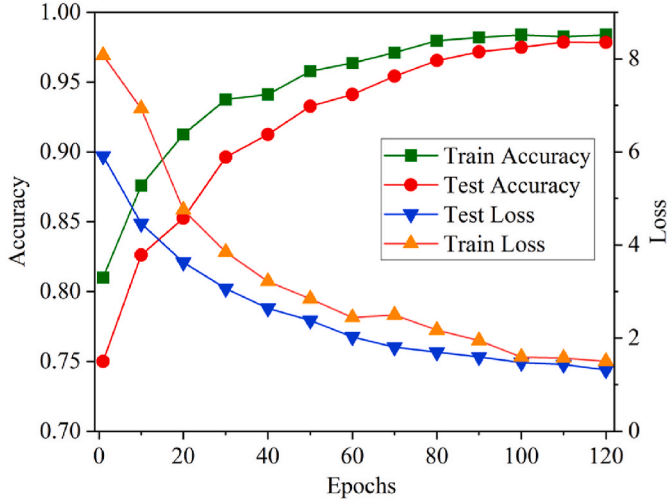
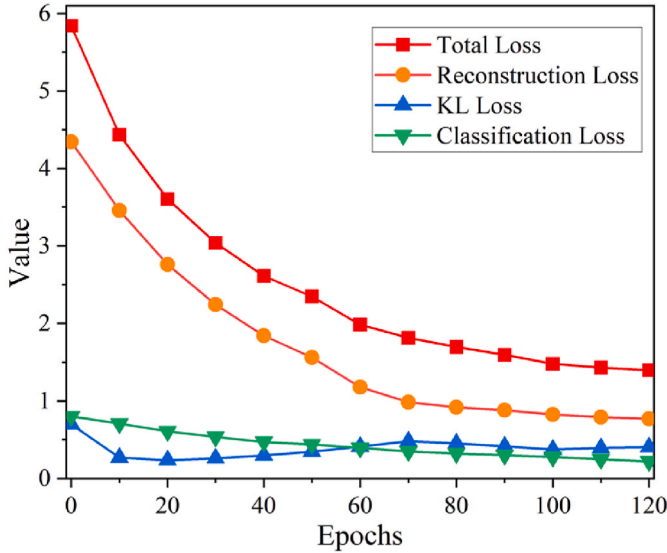
5.3. Receiver Operating Characteristic curve

The datasets of NF-UNSW-NB15-V2 and 1553B are unbalanced, with the amount of normal type exceeding that of abnormal samples, so using only accuracy as an evaluation metric cannot adequately reflect the detection ability for an unbalanced dataset. In Fig. 9, the abscissa represents the false positive rate (FPR) and the abscissa represents the true positive rate (TPR). We obtain the Receiver Operating Characteristic (ROC) curves by plotting the TPR against the FPR. For better visualization, we calculate the area under the ROC curve on the NF-UNSW-NB15-V2 and 1553B datasets in Fig. 9 (a) and (b). By comparing our results with other methods, the area value of the WKG is better. It also shows that the WKG model achieves a higher TPR and a lower FPR, indicating its robust ability to accurately identify threats while minimizing false alarms. The area results of the ROC curves show that the adaptive IDS based on the WKG model has a better performance than others in both datasets.

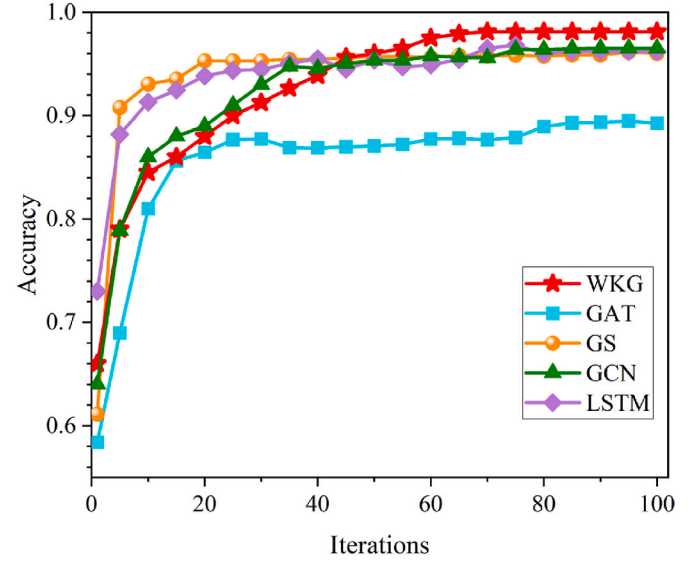
Table 2

Performance evaluation on two datasets (* indicating statistical significance).

Methods	NF-UNSW-NB15-V2				MIL-STD-1553			
	A	P	R	F1	A	P	R	F1
*SVM (Jyoti Kalita et al., 2023)	0.869	0.843	0.921	0.881	0.956	0.743	0.841	0.789
*KNN (Wenchao et al., 2014)	0.874	0.857	0.881	0.869	0.923	0.831	0.898	0.863
*LSTM	0.912	0.926	0.941	0.933	0.960	0.889	0.876	0.883
*RF (Pal, 2005)	0.927	0.934	0.916	0.924	0.862	0.818	0.857	0.837
*GCN (Gao et al., 2023)	0.907	0.788	0.676	0.726	0.951	0.749	0.753	0.751
*GAT (Velićević et al., 2017)	0.881	0.670	0.631	0.650	0.892	0.471	0.912	0.621
*GS (Mirlashari and Rizvi, 2024)	0.944	0.717	0.804	0.757	0.959	0.846	0.704	0.768
WKG	0.974	0.948	0.957	0.952	0.980	0.975	0.962	0.968

**Fig. 6.** The accuracy and loss of the WKG model.**Fig. 7.** The changes in loss components.

For imbalanced datasets, the proposed method constructs differentiated weights for data nodes using the W-KNN algorithm. Specifically, this method aggregates more class information from the k nearest neighbors for nodes near the decision boundary. For the few abnormal data nodes, the algorithm aggregates the features of multiple similar points to enhance their representativeness, thereby improving classification performance. By integrating the features of these similar nodes, the model can more accurately classify abnormal data nodes, thereby

**Fig. 8.** Accuracy comparison of different methods in 1553B.

improving intrusion detection accuracy.

5.4. Compared with the latest methods

To further prove the effectiveness of our proposed method, we compare the WKG model with state-of-the-art IDS learning methods. The performance comparison results on the NF-UNSW-NB15-V2 and 1553B datasets are shown in Tables 3 and 4.

In Table 3, we compared our method with GNN-based approaches, such as Anomal-E-IF, and ensemble methods like RF-RFE-ensemble. Anomal-E-IF, a self-supervised GNN-based approach, achieves high accuracy but performs worse than our method in F1 score and Recall. Additionally, the Anomal-E method requires integration with an anomaly detection algorithm, which limits its broader applicability. We also compared our methods with other GNN-based methods like EMA-IDS-M, DGNN, and E-GraphSAGE, which were originally proposed for Internet of Things intrusion detection. For the NF-UNSW-NB15-V2 dataset with a high degree of imbalance, the EMA-IDS-M and E-GraphSAGE methods are not as good as our method in terms of F1 and Precision metrics, while also exhibiting longer run times. Compared to the DGNN method, our method achieves a shorter runtime and demonstrates significantly superior performance in F1 score and recall, particularly in cases of higher imbalance. In addition, the GMM-WGAN-IDS and RF-RFE-ensemble integration methods require relatively long runtimes than our method, and the proposed method outperforms these methods in several metrics without the need for complex integration steps. The results highlight that our model offers a competitive advantage for network intrusion detection on the imbalanced NF-UNSW-NB15-V2 dataset. It eliminates the need for intricate integration steps,

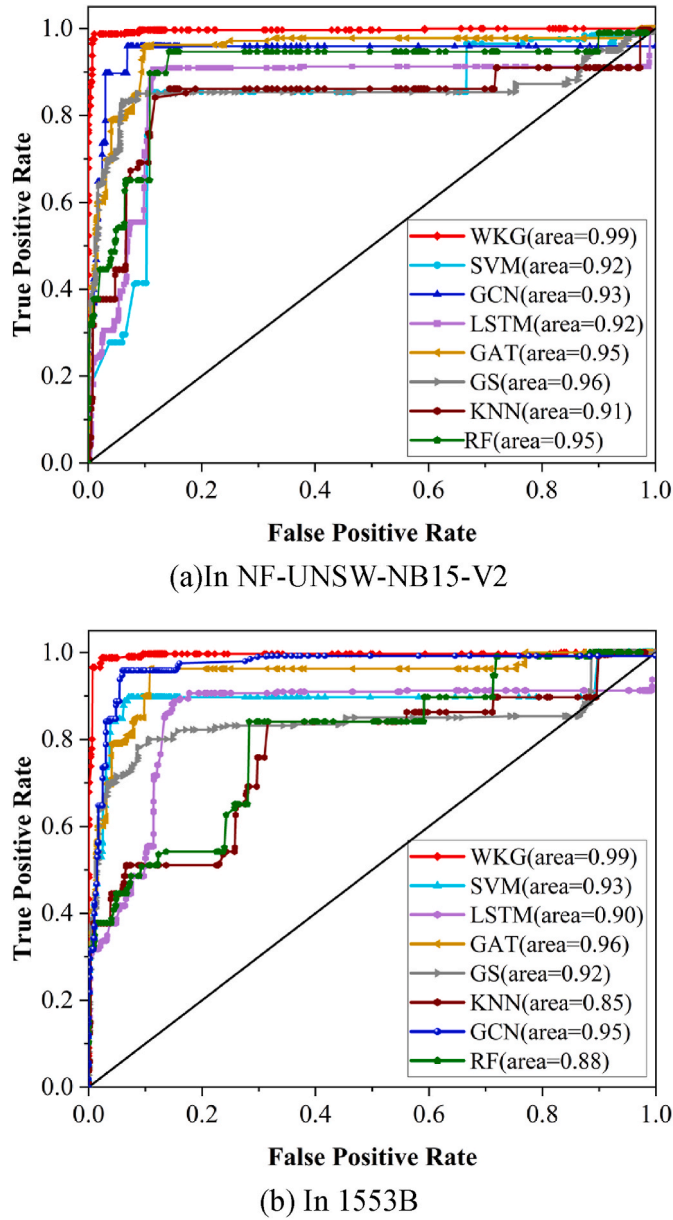


Fig. 9. ROC curves of different methods.

Table 3
Compared with the latest methods on the NF-UNSW-NB15-V2 dataset.

Method	Accuracy	F1	Recall	Precision	Time (s)
Anomal-E-IF (Caville et al., 2022)	0.979	0.856	0.687	–	62.1
EMA-IDS-M (Deng and Huang, 2025)	0.919	0.926	0.942	0.912	74.8
DGNN (Zhu et al., 2024)	0.956	0.849	0.791	0.916	92.7
E-GraphSAGE (Weng Lo et al., 2022)	0.951	0.947	0.945	0.942	77.2
GMM-WGAN-IDS (Cui et al., 2023)	0.877	0.854	0.877	0.884	–
RF-RFE-ensemble (Abbas et al., 2023)	0.899	0.862	0.868	0.862	88.3
WKG	0.974	0.952	0.957	0.948	61.7

achieves relatively shorter runtimes compared to complex ensemble learning methods, and is easy to apply in real-world applications.

The performance comparison results of the proposed model with the

Table 4

Performance comparison of the 1553B network with the latest solutions.

Method	Precision	Recall	F1	Accuracy	Time (s)
IMD-SAE-enc-dec (Huang et al., 2024)	0.847	0.921	0.905	0.975	–
OCSVM (Onodueze and Josyula, 2020)	0.579	0.001	0.001	0.903	13.1
AdaptIDS (Elsayed et al., 2022)	0.966	0.956	0.968	–	71.5
MAIDENS (Généreux et al., 2020)	0.943	0.513	0.669	–	64.0
LI-IDS (He et al., 2020)	0.925	0.783	0.861	–	46.7
OD1NF1ST (Maximilian et al., 2022)	0.953	0.757	0.844	–	43.9
WKG	0.975	0.962	0.968	0.980	16.2

latest and most relevant methods are shown in Table 4. Onodueze et al. used several machine learning models, including a one-class support vector machine (OCSVM), but their approach shows poor performance in terms of recall and F1 score than our method. This suggests that the 1553B data bus with skewed class distribution caused difficulties during training, leading to test results that were biased toward the larger proportion of normal data, which was significantly inconsistent with the actual outcomes. A similar issue is observed with the MAIDENS method, where training on only benign messages may make it hard to correctly identify anomalies. The method has a relatively long running time compared to our method. For the 1553B dataset, which is small and highly unbalanced, our method has high values for the F1 and recall metrics and is able to accurately identify anomalous data. The performance evaluation results obtained by LI-IDS are similar to those of the OD1NF1ST and IMD-SAE-enc-dec, with relatively stable performance. It achieves the recall is only 78.3 %, indicating that both models were unable to classify severely imbalanced data, which still tends to a possibility of underreporting. The AdaptIDS model has excellent intrusion performance with all performance metrics on this dataset, but has long running time compared to our method. Our method has similar performance metrics, with even higher precision. For the case of small datasets, our proposed method has clear advantages with high accuracy, less running time.

In contrast, the proposed model shows strong detection abilities for both datasets, with a high F1 score reflecting a high performance on imbalanced datasets. For airborne networks, the WKG model outperforms most existing solutions because it is easier to deploy, has shorter run times, and delivers greater accuracy.

6. Discussion

6.1. Visualization of the graph reconstruction process

In the graph reconstruction process based on the WKG model, to observe the changes in graph structure connections, we randomly selected training samples with different data sizes (3 % and 10 %) from the downsampled NF-UNSW-NB15-V2 dataset for visualization, as shown in Fig. 10. In the figure, red nodes represent attack samples, and blue nodes represent normal samples. Figures (a1) and (a2) correspond to the experiment with 3 % of the data, while figures (b1) and (b2) correspond to the experiment with 10 % of the dataset. From the figure, it can be observed that by using the Gaussian graph aggregation classification, redundant and repetitive connections are removed (red dashed lines), and the graph structure is re-aggregated based on weight similarity, adding new connections (green dashed lines). During the graph reconstruction process, the latent feature representations of nodes are obtained by sampling from a Gaussian distribution. This process removes some noise interference, resulting in improving the accuracy of the network structure.

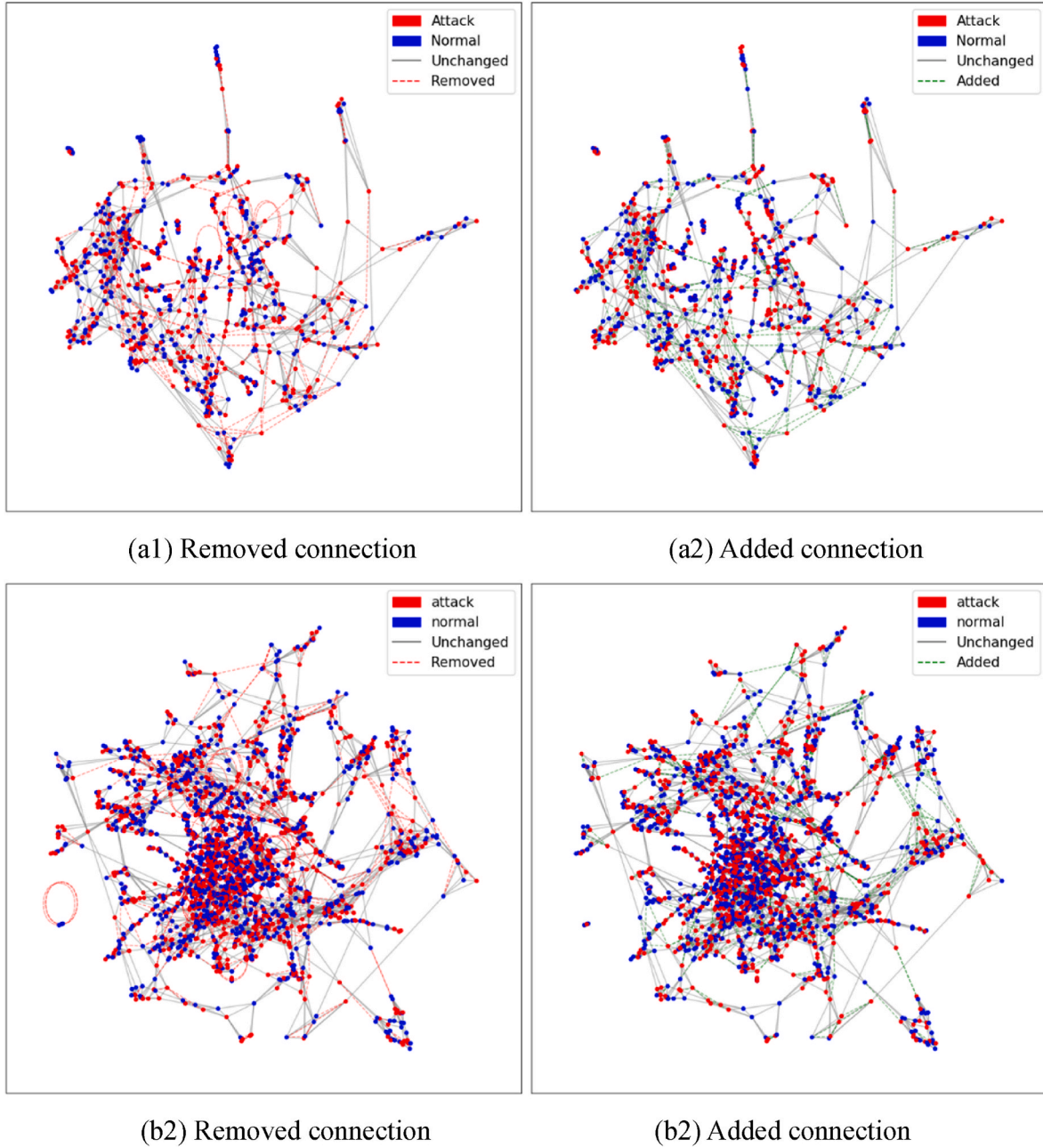


Fig. 10. Graph reconstruction process.

6.2. The learning ability in the different dataset sizes

Considering the influence of data size, we randomly selected 3 %, 5 %, 10 %, and 30 % of the original NF-UNSW-NB15-V2 dataset, respectively. We experiment with different training sizes to evaluate the influence of data size effect on our model.

In Fig. 11, we evaluate the performance of our method in accuracy, precision, recall, and F1 score. From the figure, it can be observed that as the sample size increases, the model's performance in terms of accuracy, precision, recall, and F1 score improves. There is a significant performance boost between 3 % and 5 % of the data, but from 10 % to 30 %, the model's performance remains almost unchanged. This indicates that when the sample size is between 3 % and 5 %, the model has already achieved optimal performance, and further increasing the data volume does not significantly improve the model's performance. Instead, it only increases the computational cost.

6.3. Ablation experiment

In this section, to analyze the impact of each component on the performance of the WKG model, we conducted a series of ablation experiments, and the results are summarized in Table 5. We evaluate the model based on accuracy and F1 score. The following models are compared:

To demonstrate the graph construction performance of the WKG model, we compare the KNN-based graph construction combined with graph Gaussian aggregation learning for graph reconstruction (named KG). To demonstrate the contribution of graph reconstruction to the graph structure, we compare two methods, one that combines WKNN with the GCN model (named WKNN-GCN) without the reconstruction process, and another that combines KNN with the GCN model (named KNN-GCN). Additionally, we evaluate our method against graph autoencoder methods, including Graph Variation Autoencoder (VGAE)

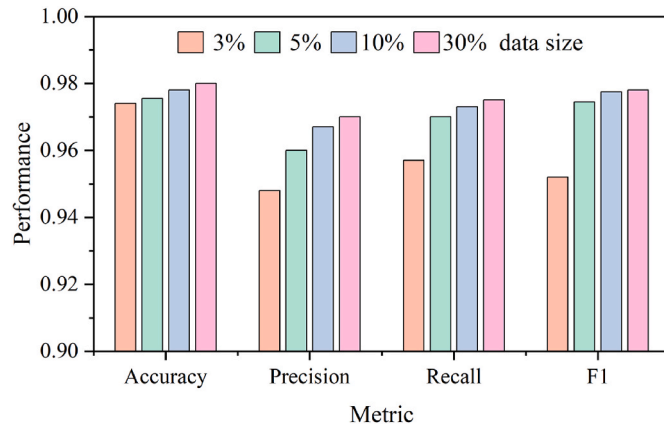


Fig. 11. Effect of training set size.

Table 5

Results of ablation experiments in NF-UNSW-NB15-V2 and 1553B dataset.

Method	NF-UNSW-NB15-V2		1553B	
	Accuracy	F1	Accuracy	F1
VGAE (Kipf and Max, 2016)	0.900	0.917	0.925	0.913
VAE (Virgile et al., 2021)	0.880	0.892	0.874	0.906
KG	0.940	0.913	0.928	0.926
WKNN-GCN	0.949	0.924	0.951	0.938
KNN-GCN	0.922	0.906	0.917	0.915
WKG($k = 3$)	0.969	0.948	0.980	0.975
WKG($k = 5$)	0.974	0.952	0.988	0.979
WKG($k = 10$)	0.975	0.950	0.982	0.970

and Graph Autoencoder (GAE). To further explore the impact of neighborhood selection during graph construction, we compare the performance of the WKG model with different numbers of k nearest neighbors.

The results in the table show that the WKNN-GCN and KNN-GCN models, which constructs the network structure using WKNN without any reconstruction technique, performs poorly on both datasets. This indicates that the reconstruction technique plays a crucial role in improving the accuracy of the network structure, as graph reconstruction serves a critical corrective function by refining initial graph constructions that may contain duplicate or redundant connections. When using similarity-based KNN for graph construction and applying graph reconstruction, the KG model exhibits a lower F1 score, suggesting that the model has not effectively addressed the imbalance within the network. The results from using the VGAE and GAE models are also unsatisfactory, with the model not achieving good performance. Additionally, we analyzed the impact of selecting the number of k nearest neighbors on the performance of the WKG model. When $k = 3$, the model connects with fewer similar nodes during the similarity-based graph construction, resulting in lower performance in both accuracy and F1 score compared to $k = 5$. This suggests that with fewer connections, the model learns from limited features and struggles to make accurate judgments for difficult nodes, leading to reduced performance. When $k = 10$, the model's performance in accuracy and F1 score remains similar. However, connecting with too many nodes leads to the model learning an excessive number of features. As the number of neighbors to connect with increases, the computational cost also rises significantly. Based on these observations, we choose $k = 5$ for the neighborhood size in this study.

7. Conclusion

In this paper, we proposed a novel WKG model, designed for airborne network intrusion detection to address the challenges of uncertainty in

graph construction and to improve the accuracy of network structure learning. The W-KNN assigns adaptive weights to samples, and by introducing differential weights for graph construction, it helps alleviate issues such as boundary point classification ambiguity and sample imbalance. The reconstruction graph of the Gaussian graph aggregation classification method focuses on the most relevant connections and features, reducing the redundant and duplicate node connections. The proposed model can not only construct the network but also continuously refine and update the network structure. By adopting the learned network structure, each sample is integrated as part of a connected group rather than as independent data items. The experimental results demonstrate that our method outperforms existing methods in some performance metrics. The proposed method can accurately detect network anomalies in airborne network datasets, offering a new perspective on intrusion detection.

In future work, we plan to develop a dynamic graph database structure. This database will dynamically update typical network attack data and related information in real time, leveraging the graph reconstruction results generated by the intrusion detection model. By continuously integrating real-time streaming data, the Gaussian distributions representing different attack types will become more refined, thereby progressively enhancing detection accuracy. Additionally, we will investigate specific dataset characteristics, such as noise and missing data, to evaluate their impact on the robustness of the WKG model and ensure its effectiveness under diverse conditions.

CRedit authorship contribution statement

Wenqi Liu: Writing – original draft. **Shijia Li:** Data curation. **Cong Gao:** Formal analysis. **Yiqin Sang:** Validation. **Hongjuan Ge:** Supervision, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be constructed as influencing the position presented in, or the review of, the manuscript entitled.

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Data availability

The authors do not have permission to share data.

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